Layer21 APP API Developer Guide

Access Point Provider (APP) Service — e-Invoicing Integration with FIRS

Layer21 Global Limited

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1. Overview

The **Layer21 Access Point Provider (APP)** API enables authorized Systems Integrators (SIs) and taxpayers to securely transmit electronic invoices to the **Federal Inland Revenue Service (FIRS)** through the Layer21 platform.

This API ensures:

- End-to-end data integrity and confidentiality.
- Compliance with FIRS e-Invoicing standards.
- Scalable and secure processing for large volumes of invoices.
- Real-time acknowledgment handling between SI → APP → FIRS → back.

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2. API Environment Setup

Environment	Base URL	Description
Production	`https://api.layer21.com/api/v1/invoice/receive`	Live FIRS-validated endpoint for production traffic.
Sandbox	`https://sandbox.layer21.com/api/v1/invoice/receive`	Layer21-hosted testing environment for ERP vendors.
Local Dev	`http://localhost:5000/api/v1/invoice/receive`	Local instance for developer testing and debugging.

All API requests must use **HTTPS** and include valid **JWT (RS256)** authorization headers.

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3. Authentication

Authorization Header

All requests must include the following header:

- **x-api-key**: 10f83d3b735a0d718047ff8febbe43b8
- **x-api-secret**: 2qr0QJz9dYzuiP7icXTY9zPUI-E4k9vd-0jh-SLCWFBR-MKy
- **Content-Type**: application/json

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4. Core API Endpoints

4.1 POST api/v1/invoice/receive

Description

Accepts invoice payloads from SI systems, validates them, and forwards to FIRS for acknowledgment.

Request Body

{

"businessId": "7b2415f1-ab22-48ba-addc-8c765c93a72c",

"irn": "INV057055532562780-2774AB22-20251021",

"issueDate": "2025-10-21",

"dueDate": "2025-10-25", //optional

"issueTime": "17:59:04", //optional

"invoiceTypeCode": "396",

"invoiceKind": "B2B",

"paymentStatus": "PENDING", //optional, defaults to pending

"note": "FIRS Demo (will be encryted in storage 2)", //optional

"taxPointDate": "2024-05-14", //optional

```
"documentCurrencyCode": "NGN",
"taxCurrencyCode": "NGN", //optional
"accountingCost": "2000", //optional
"buyerReference": "buyer REF IRN?", //optional
"invoiceDeliveryPeriod": {
    "startDate": "2024-06-14",
    "endDate": "2024-06-16"
}, //optional
"orderReference": "INV/0909/0998", //optional - Internal invoice reference
"billingReference": [
    {
        "irn": "ITW001-E9E0C0D3-20240619",
        "issueDate":"2024-05-14"
    },
    {
        "irn": "ITW001-E9E0C0D3-20240619",
        "issueDate":"2024-05-14"
    } //optional - second value to ...x in array is always optional
], //optional
"dispatchDocumentReference": {
    "irn": "ITW001-E9E0C0D3-20240619",
    "issueDate":"2024-05-14"
}, //optional
"receiptDocumentReference": {
    "irn": "ITW001-E9E0C0D3-20240619",
```

```
    "issueDate":"2024-05-14"
  }, //optional
  "originatorDocumentReference": {
    "irn": "ITW001-E9E0C0D3-20240619",
    "issueDate":"2024-05-14"
  }, // optional
  "contractDocumentReference": {
    "irn": "ITW001-E9E0C0D3-20240619",
    "issueDate":"2024-05-14"
  }, //optional
  "documentReference": [
    {
      "irn": "ITW001-E9E0C0D3-20240619",
      "issueDate":"2024-05-14"
    }
  ], //optional
  "accountingSupplierParty": {
    "partyName": "Test Pls", //optional if id is set
    "tin": "76819548-6539",
    "email": "supplier_business@email.com",
    "telephone": "+23480254099000", //optional, must start with + (meaning country code)
    "businessDescription": "this entity is into sales of Cement and building materials",
//optional
    "postalAddress": {
      "streetName": "32, owonikoko street", //optional if id is set
      "cityName": "Gwarikpa", //optional if id is set
```

```
        "postalZone": "023401", //optional if id is set
        "country": "NG" //optional if id is set
      } //optional if id is set
   },
   "accountingCustomerParty": {
      "partyName": "Test Pls", //optional if id is set
      "tin": "76819548-6539",
      "email": "business@email.com",
      "telephone": "+23480254000000", //optional, must start with + (meaning country code)
      "businessDescription": "this entity is into sales of Cement and building materials",
//optional
      "postalAddress": {
        "streetName": "30, owonikoko street", //optional if id is set
        "cityName": "Gwarikpa", //optional if id is set
        "postalZone": "023401", //optional if id is set
        "country": "NG" //optional if id is set
      } //optional if id is set
   },
   // "payeeParty": {}, //optional (party object, just like accounting_customer_party)
   // "taxRepresentativeParty": {}, //optional (party object, just like
accounting_customer_party)
   "actualDeliveryDate": "2024-05-14", //optional
   "paymentMeans": [
      {
        "paymentMeansCode": "10",
        "paymentDueDate": "2024-05-14"
```

```
    },
    {
        "paymentMeansCode": "43",
        "paymentDueDate": "2024-05-14"
    }//optional - second value to ...x in array is always optional
  ],//optional
  "paymentTermsNote": "Just a test payment (will be encryted in storage)",//optional
  "allowanceCharge": [
    {
        "chargeIndicator": true, //indicates whether the amount is a charge (true) or an
allowance (false)
        "amount": 800.60
    },
    {
        "chargeIndicator": false, //indicates whether the amount is a charge (true) or an
allowance (false)
        "amount": 10
    }//optional - second value to ...x in array is always optional
  ],//optional
  "taxTotal": [
    {
        "taxAmount": 56.07,
        "taxSubtotal": [
            {
                "taxableAmount": 800,
                "taxAmount": 8,
```

```
            "taxCategory": {
                "id": "LOCAL_SALES_TAX",
                "percent": 2.3
            }
        }
    ]
  }//second value to ...x in array is always optional if you want to add it
],
"legalMonetaryTotal": {
  "lineExtensionAmount": 340.50,
  "taxExclusiveAmount": 400,
  "taxInclusiveAmount": 430,
  "payableAmount": 30
},
"invoiceLine": [
  {
    "hsnCode": "8471.49", //
    "productCategory": "Automatic data processing machines", //
    "discountRate": 2.01, //
    "discountAmount": 3500, //
    "feeRate": 1.01, //
    "feeAmount": 50, //
    "invoicedQuantity": 15,
    "lineExtensionAmount": 30,
    "item": {
```

```
            "name": "item name",
            "description": "item description",
            "sellersItemIdentification": "identified as spoon by the seller" //optional
        },
        "price": {
            "priceAmount": 10,
            "baseQuantity": 3,
            "priceUnit": "NGN per 1"
        }
    },

    }//optional - second value to ...x in array is always optional
  ]
}
```

Response

```
{
  "status": "success",
  "firsAckId": "ACK-123456789",
  "validationStatus": "transmitted",
  "timestamp": "2025-10-11T10:22:58Z"
}
```

Error Codes

Code	Message	Meaning
400	INVALID_SCHEMA	JSON structure or field missing
401	INVALID_SIGNATURE	JWT validation failed
422	COMPLIANCE_ERROR	VAT or TIN validation failed
500	FIRS_ENDPOINT_ERROR	FIRS endpoint unreachable

4.2 GET /v1/invoices/status/{invoiceId}

Description

Retrieves validation and acknowledgment status of a submitted invoice.

Response Example

```
{
  "invoiceId": "INV-2025-00023",
  "status": "transmitted",
  "firsAckId": "ACK-123456789",
  "validationTimestamp": "2025-10-11T10:22:58Z"
}
```

4.3 POST /v1/firs/ack

Description

Receives acknowledgment notifications from FIRS and updates invoice status.

Example Payload

```
{
  "firsAckId": "ACK-123456789",
  "invoiceId": "INV-2025-00023",
  "status": "transmitted",
  "remarks": "Signed successfully",
  "timestamp": "2025-10-11T10:25:00Z"
}
```

Expected Response

```
{ "status": "acknowledged" }
```

4.4 GET /v1/healthz

Description

Simple health check endpoint to verify API availability.
Response: `Self-health-check result: {'ok': True, 'response': {'code': 200, 'data': {'ok': True, 'item': {'dummy_test_invoice_notification_sent': True, 'health_check_completed': True, 'new_setup': False}}}}`

4.5 GET /v1/metrics

Description

Prometheus-compatible metrics for system monitoring and performance tracking.

Sample Output

```
app_requests_total{status="200"} 23456
app_requests_failed{status="500"} 32
```

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5. Security & Compliance

Layer21 implements the following controls:

- **Encryption:** All API traffic uses TLS 1.3.
- **Authentication:** RS256 JWT signatures, verified per request.
- **Authorization:** Role-based access (Integrator, Admin, Auditor).
- **Data Storage:** Encrypted PostgreSQL database (AES-256 at rest).
- **Audit Trail:** Immutable logs for every transaction.
- **Rate Limiting:** 200 requests/min per SI key.
- **Monitoring:** Real-time Prometheus & Grafana dashboards.
- **Compliance:** NDPR, FIRS MBS, and NITDA Cybersecurity Guidelines.

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6. Developer Tools

Testing Environment

Developers can use the sandbox environment for testing:
`https://sandbox.layer21.com/app`

Sandbox supports simulated FIRS acknowledgments and sample payloads.

Sample Test JWT

```
export TOKEN=$(curl -X POST https://sandbox.layer21.com/app/auth/test-
token)
curl -X POST https://sandbox.layer21.com/app/v1/invoices/submit \
  -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d @invoice.json
```

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7. Versioning & Updates

Version	Status	Notes
v1.0	Active	Initial release for SI-to-FIRS integration
v1.1	Planned	Bulk invoice submission & webhook retries

All API changes are backward compatible. Clients will receive 30-day notice before any version deprecation.

8. Support & Contact

Department	Email	Availability
Technical Support	support@layer21.com	24/7
Developer Relations	dev@layer21.com	Mon–Fri, 9AM–6PM
Incident Escalation	ops@layer21.com	Immediate response (P1 issues)

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9. Legal & Compliance

All API usage is governed by:

- FIRS e-Invoicing compliance framework.
- Layer21's internal Information Security Management Policy.
- Nigeria Data Protection Regulation (NDPR 2019).

Misuse of access credentials or unauthorized data processing constitutes a breach under Section 37 of the Nigerian Constitution and NDPR enforcement guidelines.

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10. Appendix: API Flow Diagram (Text Description)

Flow:
`Taxpayer ERP → Layer21 SI → Layer21 APP → FIRS Gateway → FIRS ACK → APP → SI → ERP`

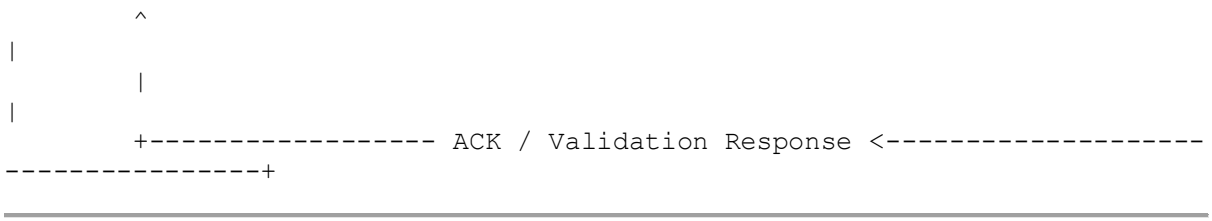
Each stage includes validation, digital signing, and secure message forwarding.

Appendix B — API Flow Diagram & Legend

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1. Text-Based Flow Diagram

```
+-------------------+        +-----------------+        +-------------------+
+------------------+
|   Taxpayer ERP    | --->   | Layer21 SI API | --->   | Layer21 APP API   |
--->   |   FIRS Gateway   |
+-------------------+        +-----------------+        +-------------------+
+-----------------+
```



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2. Flow Description

Step	Actor	Action Description	Security Controls
1	**Taxpayer ERP**	Generates electronic invoice in JSON format.	Local signing with client private key (optional).
2	**Layer21 SI**	Validates JSON schema and applies RS256 digital signature.	JWT signing, payload validation, and logging.
3	**Layer21 APP**	Receives signed invoice, runs compliance checks, and forwards to FIRS.	TLS 1.3 encryption, token verification, and schema validation.
4	**FIRS Gateway**	Validates invoice and issues acknowledgment (ACK).	FIRS digital signature, reference ID generation.
5	**Layer21 APP → SI → ERP**	Routes acknowledgment back to client ERP.	End-to-end integrity validation using signature comparison.

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3. Legend

Icon / Symbol	Meaning
□	Invoice payload (JSON schema)
🔏	RS256 digital signature
🌐	HTTPS/TLS encrypted transport
⚙️	Validation and compliance logic
📨	ACK (Acknowledgment) response
📊	Metrics and observability layer
□	PostgreSQL data persistence (encrypted at rest)

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4. Security Layers Overview

1. **Network Security:**
   - Enforced HTTPS connections (TLS 1.3 only).
   - Strict HSTS and Content Security Policy headers.
2. **Application Security:**
   - JWT tokens signed with RS256 private key.
   - Payloads validated against official FIRS JSON schemas.
3. **Data Security:**
   - AES-256 database encryption at rest.

- Daily encrypted backups to Nigerian data centers.

4. **Operational Security:**
   - Continuous monitoring via Prometheus and Grafana.
   - Incident response window ≤ 15 minutes (P1 events).

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5. Summary Visualization (for reference)

Use this brief description when creating a visual diagram slide for presentations:

"Invoices generated by taxpayer ERPs are securely transmitted to the **Layer21 System Integrator (SI)**, validated, and signed before being sent to the **Layer21 Access Point Provider (APP)**.
The APP applies compliance checks and forwards them to the **FIRS Gateway**, which returns an acknowledgment (ACK).
This ACK is then routed back through APP → SI → ERP in real time, maintaining a complete audit trail."

Footer (for all pages):

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